

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

QIAN QI
Peking University

This supplement provides the operator proofs, verification conditions, transition construction, additional numerical results, and replication instructions for the seventh revision. It distinguishes continuous-time economic objects from finite transition approximations and distinguishes trained neural policies from analytical and finite-state regression tests. Additional proofs characterize localized count dominance, compressed work, finite refinement, primitive CRRA option bounds, and the error account for a joint-region risk certificate. The main paper and this supplement use the same policy-evaluation, boundary, and temporal-self conventions.

S.1. OPERATOR PROOFS AND QUANTITATIVE APPROXIMATION

S.1.1. *Exact Policy Improvement*

We use the assumptions of the main paper’s exact separated-operator proposition. In particular, $I(p)(t, s)$ is a selector maximizing $\mathcal{H}^a[E(p)](t, s)$, not a stationary point of a parameterized actor objective. Continuity of E is in the norm controlling the value, time derivative, first spatial derivatives, second spatial derivatives, and the stated boundary traces. The admissible compact policy class is closed under I . These are assumptions on an exact boundary-value problem, not universal properties of neural policy classes.

PROOF OF EXACT SEPARATED-OPERATOR CONSISTENCY: Write $V_n = E(p_n)$ and $p_{n+1} = I(p_n)$. The evaluation equation gives $V_{n,t} + \mathcal{H}^{p_n}[V_n] = 0$. Pointwise maximization implies

$$V_{n,t} + \mathcal{H}^{p_{n+1}}[V_n] \geq 0.$$

The assumed comparison and verification inequalities, with the same boundary operator for both policies, imply $V_{n+1} \geq V_n$. Precompactness of the evaluation image implies boundedness and hence a pointwise limit $\bar{V}$. Every subsequential limit in the declared strong topology has this same value component. Its derivative components are the derivatives of that value, because uniform convergence of the functions and their derivatives preserves differentiation. Consequently all strong accumulation points coincide and $V_n \rightarrow \bar{V}$ strongly.

Let $p_{n_j} \rightarrow p$ be a policy subsequence, which exists by compactness. Continuity of evaluation gives $E(p) = \bar{V}$. Continuity of improvement at p and continuity of evaluation give

$$E(I(p)) = \lim_j E(I(p_{n_j})) = \lim_j V_{n_j+1} = \bar{V}.$$

Evaluating $I(p)$ at this common value yields

$$0 = \bar{V}_t + \mathcal{H}^{I(p)}[\bar{V}] = \bar{V}_t + \sup_a \mathcal{H}^a[\bar{V}].$$

The boundary traces also pass to the limit. Comparison identifies $\bar{V}$ with the optimal value, and verification identifies p as optimal because $E(p) = \bar{V}$. This argument does not require $p = I(p)$ or convergence of the entire policy sequence. In the presence of tied maximizing actions, different optimal policies may share the same limiting value. Q.E.D.

The proof is an application of exact policy-improvement reasoning; it is not a claim of priority over [Jacka and Mijatović \(2015\)](#). The assumption of a compact strongly regular evaluation image can fail at corners or under degenerate diffusions. The stopped application is therefore evaluated by its specified positive transition approximation rather than asserted to satisfy every classical hypothesis globally.

S.1.2. Finite-Horizon Value-Loss Bound

For clarity, put $\Gamma_{n,n} = 1$ and $\Gamma_{n,j} = \prod_{\ell=n}^{j-1} \gamma_\ell$. Both $\mathcal{T}_n^a$ and its policy version are monotone and γ_n -Lipschitz in the sup norm. The implemented operator is $\tilde{\mathcal{T}}_n^a$. The three nonnegative error bounds at $\hat{v}_{n+1}$ are

$$\begin{aligned}\epsilon_n &\geq \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \delta_n &\geq \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \kappa_n &\geq \sup_a \left\| (\tilde{\mathcal{T}}_n^a - \mathcal{T}_n^a) \hat{v}_{n+1} \right\|_\infty.\end{aligned}$$

Here $\tilde{\mathcal{T}}_n = \sup_a \tilde{\mathcal{T}}_n^a$, and $\|\hat{v}_N - G\|_\infty \leq \eta$.

PROOF OF THE MAIN VALUE-LOSS THEOREM: Let $D_n = \left\| \hat{v}_n - J_n^{\hat{p}} \right\|_\infty$. Inserting the implemented and target fixed-policy backups gives

$$D_n \leq \epsilon_n + \kappa_n + \gamma_n D_{n+1}.$$

Let $R_n = \|\hat{v}_n - V_n^*\|_\infty$. The supremum inequality $|\sup_a f_a - \sup_a g_a| \leq \sup_a |f_a - g_a|$ and the nonnegative improvement gap give

$$R_n \leq \epsilon_n + \delta_n + \kappa_n + \gamma_n R_{n+1}.$$

Backward iteration, with $D_N, R_N \leq \eta$, proves

$$\begin{aligned}D_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta, \\ R_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \delta_j + \kappa_j) + \Gamma_{n,N} \eta.\end{aligned}$$

Feasibility implies $V_n^* \geq J_n^{\hat{p}}$. Combining the two inequalities proves the stated upper bound with coefficients $2\epsilon_j + \delta_j + 2\kappa_j$ and terminal coefficient 2η . Q.E.D.

The theorem is finite-horizon. It permits a recursive operator with $\gamma_n > 1$ but then records amplification explicitly through $\Gamma_{n,j}$. It does not imply contraction for an arbitrary Epstein–Zin aggregator. Boundary killing can be included in the transition kernel by treating the boundary payoff as a known reward component; only continuation mass contributes to γ_n .

S.1.3. Differential Residual and Boundary Error

Let v be a $C^{1,2}$ candidate for the additive, stopped, discounted problem. Define $R^a = v_t + \ell^a + \mathcal{L}^a v - \rho v$. Suppose $|R^{\hat{p}}| \leq \epsilon(t)$, $\sup_a R^a - R^{\hat{p}} \leq \delta(t)$, and the payoff discrepancy at terminal time or exit is at most η . Assume the localized Itô formula passes to expectations by uniform integrability.

For every admissible p , stopped Itô integration gives

$$J^p(t, s) - v(t, s) = \mathbb{E}_{t,s}^p \left[\int_t^\tau e^{-\rho(r-t)} R^{p_r}(r, S_r) dr + e^{-\rho(\tau-t)} (G - v)(\tau, S_\tau) \right].$$

The upper bound for an arbitrary policy is $\eta + \int_t^T e^{-\rho(r-t)} (\epsilon(r) + \delta(r)) dr$. The absolute discrepancy for $\hat{p}$ is at most $\eta + \int_t^T e^{-\rho(r-t)} \epsilon(r) dr$. Taking a supremum over p and adding proves the differential corollary. The argument is conditional on uniform residual and boundary bounds. Replacing these bounds by sampled root mean squared losses is not a valid proof step.

S.1.4. Feasible Candidates, Convex Gaps, and Reference Bounds

If $Q(s, a)$ is L_a -Lipschitz in a and a feasible set $\mathcal{C}(s)$ is an h_a -net of the admissible actions, then a maximizer over $\mathcal{C}(s)$ is within $L_a h_a$ of the full supremum. Adding a neural proposal to $\mathcal{C}(s)$ cannot reduce its achieved backup value. Neither statement bounds the evaluation error of the critic that generated Q .

For differentiable convex F on a compact convex set A , its tangent inequality implies

$$F(a) - \inf_{b \in A} F(b) \leq \nabla F(a)^\top a - \min_{b \in A} \nabla F(a)^\top b =: g_F(a).$$

For $A = \{a \geq 0 : \sum_i a_i \leq B\}$, the minimum is $B \min(0, \min_i \partial_i F(a))$. The same formula applies to a scenario tree with separate feasible actions at each information node: sum the node-specific linear minimization gaps, using the gradient of the complete probability-weighted objective. This gradient must include all descendant costs.

If a feasible reference has objective C_R and gap G_R , then $C_R - G_R \leq C^* \leq C_R$. Any other feasible policy, evaluated on the same tree, therefore satisfies

$$C_\pi - C_R \leq C_\pi - C^* \leq C_\pi - C_R + G_R.$$

The program evaluates both objectives by all 64 terminal branches. It does not insert the neural critic as a substitute for policy cost. The reported interval is based on ordinary double-precision evaluation of these inequalities; it is not an interval-arithmetic enclosure of rounding error.

A useful conditional policy bound also follows. If the optimal continuation backup $Q^*(s, \cdot)$ is μ -strongly concave and the approximate continuation is within e of the

optimal one, an action whose approximate improvement gap is at most δ obeys

$$\|\hat{a} - a^*\|^2 \leq \frac{2}{\mu}(\delta + 2\gamma e).$$

Indeed the value of each action changes by at most γe , and strong concavity bounds the resulting true suboptimality below by $\mu \|\hat{a} - a^*\|^2 / 2$. This is not applied to a nonconcave NDU maximand. In the convex resource problem the immediate quadratic cost supplies strong convexity $0.2/d$ for minimization. Its deterioration with dimension is included in the conditioning discussion rather than hidden in a dimension-free speed claim.

S.2. COUNTEREXAMPLES, TRACE ESTIMATION, AND DIFFERENTIATION

S.2.1. *Why a Joint Penalty Does Not Define the Economic Solution*

Consider $-V' + V = 0$ on $[0, 1]$, $V(1) = 1$. Its solution is $V^*(t) = e^{t-1}$. The historical composite objective was of the form

$$L(V) = \int_0^1 \{V(t) + [-V'(t) + V(t)]^2\} dt.$$

The feasible perturbation $V_\varepsilon(t) = V^*(t) - \varepsilon(1 - t)$ preserves the terminal condition but has residual $-\varepsilon(2 - t)$. Consequently

$$L(V_\varepsilon) - L(V^*) = -\frac{\varepsilon}{2} + \frac{7\varepsilon^2}{3} < 0 \quad \text{for } 0 < \varepsilon < \frac{3}{14}.$$

Thus the exact economic solution is not even a local minimizer of this penalty. Separate parameter blocks, objectives, and verification conditions are substantive changes, not a change of notation for that joint loss.

S.2.2. *A Stable Actor Stationary Point Need Not Be Optimal*

On $[-2, 2]$, let $H(a) = -(a^2 - 1)^2 + 0.2a$. Its stationary equation is $4a^3 - 4a - 0.2 = 0$. There are three real roots. At the negative local maximizer, approximately -0.9739943532 , $H''(a) = 4 - 12a^2 < 0$, so gradient ascent has a locally asymptotically stable equilibrium. The positive maximizer is approximately 1.0241203002 and has higher payoff by approximately 0.3998745872 . Both are isolated. The example embeds into a finite-horizon control problem with flow $H(a)$ and exact evaluation $(T - t)H(a)$.

Consequently, even exact fast evaluation, diminishing separated step sizes, bounded iterates, and zero noise do not turn isolated actor stationary points into policy-improvement fixed points. Under a suitable stochastic-approximation theorem the justified conclusion concerns the limiting invariant or stationary set. Global optimality additionally requires an argument excluding suboptimal attractors. Convex improvement in the resource example supplies such an argument for each surrogate action problem; it does not automatically eliminate critic approximation error. Constant-step Adam has no convergence theorem in this paper, and the executed neural runs use the explicitly recorded L-BFGS and constrained least-squares routines instead.

S.2.3. Small Strong Residuals Can Miss a Viscosity Solution

For the exit-time problem $dX_t = a_t dt$, $|a_t| \leq 1$, with flow -1 and zero payoff on $\{-1, 1\}$, the value is $V(x) = |x| - 1$. The viscosity equation is $1 - |V'| = 0$. The smooth functions

$$W_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2}$$

satisfy the boundary data, while their squared strong residual under the uniform distribution tends to zero:

$$\frac{1}{2} \int_{-1}^1 \left(1 - \frac{|x|}{\sqrt{x^2 + \varepsilon^2}} \right)^2 dx \sim \left(2 - \frac{\pi}{2} \right) \varepsilon.$$

Nevertheless $W_\varepsilon \rightarrow 1 - |x|$, which has the wrong sign and is not the desired viscosity solution. The asymptotic expression follows by $x = \varepsilon y$ and integration of the resulting integrable function on $[0, \infty)$. The code also evaluates the finite- ε integral by independent adaptive quadrature.

Monotonicity, stability, and consistency with an appropriate comparison principle are central to viscosity approximation; see [Barles and Souganidis \(1991\)](#). Smooth network approximation of a function and its derivatives, as in [Hornik, Stinchcombe, and White \(1990\)](#), concerns a smooth target and does not approximate a derivative at a kink or replace a viscosity-selection argument. Historical kink and residual arrays remain available as archival material, not as evidence that an unregularized strong-residual loss selects the correct solution.

S.2.4. Probe Budget and the Squared-Loss Estimand

Let $A = B^\top \nabla^2 v B$ be symmetric, z_k independent standard Gaussian vectors, and $q_k = z_k^\top A z_k$. Gaussian fourth moments give $\mathbb{E} q_k = \text{tr } A$ and $\text{Var}(q_k) = 2 \|A\|_F^2$. For deterministic a and $\bar{q}_K = K^{-1} \sum_k q_k$,

$$\mathbb{E} \left(a - \frac{1}{2} \bar{q}_K \right)^2 = \left(a - \frac{1}{2} \text{tr } A \right)^2 + \frac{\|A\|_F^2}{2K}.$$

In contrast $K^{-1} \sum_k (a - q_k/2)^2$ has excess expectation $\|A\|_F^2/2$, independent of K . For the regression matrix $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\|A\|_F^2 = 15$, $a = 1.25$, and the exact-trace loss is 1.5625. The population excess is $7.5/K$.

An unbiased alternative for $K \geq 2$ is the off-diagonal statistic

$$U_K = \frac{(\sum_k r_k)^2 - \sum_k r_k^2}{K(K-1)}, \quad r_k = a - q_k/2.$$

Independence implies $\mathbb{E} U_K = (\mathbb{E} r_k)^2$. Under an integrable derivative envelope, differentiating its expectation yields the gradient of the exact squared residual. The realized statistic can be negative; unbiasedness neither makes it a nonnegative sample loss nor guarantees optimizer stability. If the action or parameters are selected using the same probes, the fixed-parameter identity is not an adaptive generalization theorem. Independent held-out probes are needed to interpret such selection.

TABLE S.1
INDEPENDENT-BATCH STOCHASTIC-TRACE LOSSES

K	Correct mean	Population	MC s.e.	Old mean	Debiased mean
1	8.91810	9.06250	0.22234	8.91810	–
4	3.47199	3.43750	0.04758	9.17095	1.57233
16	2.03146	2.03125	0.01492	9.06287	1.56270
64	1.68023	1.67969	0.00640	9.06132	1.56307

20,000 independent Gaussian batches per row; the exact-trace squared residual is 1.5625. Correct mean averages probes before squaring; old mean averages individual squared residuals. The debiased off-diagonal statistic requires at least two probes. Its standard errors and both variance estimands are retained in the JSON record.

Each row averages 20,000 independent batches. The Monte Carlo standard error pertains to the repeated-batch mean, not to a single quadratic form or its within-batch variance. The numerical test checks the known expectation within six Monte Carlo standard errors and preserves the wrong mean-of-squares statistic as a negative control. It does not assign value or policy errors to this algebraic experiment.

S.2.5. Executable Computational-Graph Tests

The test suite forms a quadratic critic and a control-dependent diffusion loading, contracts its automatic Hessian with the covariance, and checks both parameter blocks. In the actor update, critic coefficients receive no gradient but the action-dependent covariance does. In the critic update, the policy output is detached while the critic coefficients receive gradients. A finite-transition actor derivative is also checked against central differences in all three controls. Freezing critic parameters does not detach the next-state input: the continuation gradient with respect to the action-induced state is retained.

A separate test compares the resource critic’s explicit gradient with PyTorch automatic differentiation. The convex squared-ReLU feature architecture is a transparent special case of the input-convex approach of [Amos, Xu, and Kolter \(2017\)](#). Exact directional second derivatives or Hessian-vector products, following [Pearlmutter \(1994\)](#), can avoid constructing a dense Hessian when the diffusion has low rank. The complete costs still depend on network size, diffusion multiplication, sample count, probe count, and optimization iterations.

S.3. THE STOPPED PREFERENCE-FORMATION ECONOMY

S.3.1. Admissibility and the Boundary Contract

The interior state is $(u, X) \in (1.2, 2.8) \times (0.5, 2)$, with the controls and coefficients stated in the main paper. The process is stopped at the first exit or $T = 1$, whichever occurs first. It receives $G(u, X) = -0.02(u - 2)^2 + 0.1 \log X$ at that time. A start on the boundary is settled immediately. This is a finite-horizon contract on an open operating domain; it is not a claim that the unstopped diffusion remains in a rectangle.

For bounded progressively measurable controls, the wealth SDE with coefficients linear in wealth and the additive preference SDE have a well-defined solution up to exit. Before exit, wealth and consumption are bounded away from zero. The flow, terminal payoff, and quadratic adjustment cost are bounded on the stopped domain. Thus the

additive objective is finite without introducing a reflected local time or a wealth-transfer process.

The original state-constraint interpretation cannot hold with these coefficients. Preference volatility is 0.05 at both endpoints, and the maximal wealth drift at the lower boundary is $(0.02 + 0.8 \times 0.06)0.5 - 0.05 = -0.016$. At zero portfolio volatility it is more negative. Neither fact is repaired by projecting an Euler endpoint back onto the rectangle. The stopped contract retains all interior coefficients and makes the change in boundary economics explicit.

S.3.2. A Boundary-Compatible Payoff

The choice of G has a useful analytic check. Its derivatives are $G_u = -0.04(u - 2)$, $G_{uu} = -0.04$, $G_X = 0.1/X$, $G_{XX} = -0.1/X^2$, and $G_{uX} = 0$. For every admissible action and every $k \geq 0$,

$$\begin{aligned} \ell^a + \mathcal{L}^a G - \rho G &\leq \max_{u \in [1.2, 2.8]} \frac{0.8^{1-u}}{1-u} + 0.0064 + 0.0043 - 0.00005 + 0.0032846 \\ &< -0.8162. \end{aligned}$$

The consumption bound follows because utility is increasing in consumption. Maximization over u at $c = 0.8$ gives the upper endpoint; it can also be checked by the explicit derivative. The 0.0043 term bounds $0.1[0.02 + 0.06\pi - c/X]$ by using $\pi \leq 0.8$, $c/X \geq 0.025$. The omitted portfolio diffusion term is nonpositive, and so are adjustment costs. The last term bounds $-\rho G$ uniformly.

Stopped Itô integration therefore implies $J^p \leq G$ for every admissible policy. To obtain boundary continuity from below, fix for example a constant admissible nonzero portfolio share. Its two-dimensional covariance is uniformly positive definite on the closed rectangle because $X \geq 0.5$ and $|\varrho| < 1$. The boundary is regular for this fixed diffusion, and its expected exit time tends to zero as the start tends to a boundary point. Bounded rewards and continuity of G imply that this policy's stopped payoff tends to G there. Since its payoff is bounded above by the optimal value and the latter is bounded above by G , the value has the declared boundary limit. This argument establishes boundary compatibility; it does not assert global classical differentiability at corners.

S.3.3. Transition, Covariance, and First Exit

For step h , form four equiprobable sign pairs $(z_1, z_2) \in \{-1, 1\}^2$. Let

$$\begin{aligned} d_u &= \theta h + \sigma_u \sqrt{h} z_1, \\ d_X &= [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} [\varrho z_1 + \sqrt{1 - \varrho^2} z_2]. \end{aligned}$$

For an interior state s , α is the first fraction in $[0, 1]$ at which the line $s + \alpha d$ meets a face, or one if it stays inside. An exit branch has payoff

$$\frac{1 - e^{-\rho \alpha h}}{\rho} \ell(s, \alpha) + e^{-\rho \alpha h} G(s + \alpha d).$$

A surviving branch instead uses the continuation value at $s + d$ and the full-step flow annuity. At a boundary start the backup is exactly $G(s)$. In particular there is no nonzero continuation from a boundary node. The reference evaluator uses nonnegative bilinear interpolation weights that sum to one on surviving branches.

These signs reproduce conditional first moments and the covariance matrix of the written diffusion. Polynomial tests on $1, u, X, u^2, X^2, uX$ distinguish covariance terms from drift-product terms of order h . At the test state and action the covariance is

$$\begin{pmatrix} 0.0025 & -0.002 \\ -0.002 & 0.0256 \end{pmatrix}.$$

The largest generator discrepancy decreases by a factor of ten when h decreases from 10^{-2} to 10^{-3} and again to 10^{-4} . Preference controls of -0.2 and 0.2 give different mean next states even when both displacements are smaller than a spatial grid cell. No nearest-neighbor map erases the control.

The line-exit rule is an approximation to diffusion killing, not exact Brownian bridge detection. Before-correction overshoots, branch-exit frequencies, occupied exit probabilities, and post-exit payoffs are distinct recorded objects. A reflection-regulator field is inapplicable and is represented by null with that explanation. For the joint state–time refinement, the squared largest mesh width divided by h decreases as $0.04, 0.02, 0.01$; the separate control refinements are also retained. These observations support consistency diagnostics but do not supply a numerical bound for every κ_n in the main theorem.

S.3.4. *Economic Comparative Statics*

For each admissible policy let $B(p)$ denote cost-free benefit. Define discounted cumulative effort by $C(p) = \mathbb{E} \int_0^{\tau_p} e^{-\rho t} \theta_t^2 / 2 dt$. The stopping time is policy dependent but not directly dependent on k for a fixed policy. Thus $V(k) = \sup_p [B(p) - kC(p)]$ is decreasing and convex. For optimizers p_1, p_2 at $k_1 < k_2$, revealed optimality gives

$$\begin{aligned} B(p_1) - k_1 C(p_1) &\geq B(p_2) - k_1 C(p_2), \\ B(p_2) - k_2 C(p_2) &\geq B(p_1) - k_2 C(p_1). \end{aligned}$$

Adding implies $(k_2 - k_1)[C(p_2) - C(p_1)] \leq 0$. Also $-C(p)$ is a subgradient of V at any k for which p is optimal, and $V'(k) = -C(p)$ wherever V is differentiable. For policies with value losses at most e_1, e_2 , the same inequalities with these errors imply

$$C(p_2) - C(p_1) \leq \frac{e_1 + e_2}{k_2 - k_1}.$$

The result applies to the continuous stopped model and separately to each exactly specified positive finite approximation. A finite-grid comparison cannot set $e_i = 0$ for the continuous problem merely because its discrete maximization was exhaustive.

The reported effort is computed both backward and by forwarding the induced state probabilities. The latter multiplies live branch mass by bilinear weights and accumulates discounted cost through the line-exit fraction. This is an occupation calculation for the interpolation-defined finite Markov chain. It provides an economically interpretable sum rather than an unweighted mean of control magnitudes. A reduction of this aggregate does not imply a pointwise reduction of θ , consumption, or portfolio share. Nor does a comparative static identify k from observations without a specified data model and an injective moment map.

S.4. PORTFOLIO AND RECURSIVE-UTILITY VERIFICATION

S.4.1. *Stationary Merton Benchmark*

Write $b = 1 - \gamma$, $V(X) = AX^b/b$, $c = mX$, and

$$g(m, \pi) = r + (\mu - r)\pi - m + \frac{1}{2}(b - 1)\sigma^2\pi^2.$$

For exponential utility discount ρ , the fixed-policy stationary equation is

$$A = \frac{m^b}{\rho - bg(m, \pi)}$$

on the positive branch $\rho - bg > 0$. Optimization gives $\pi^* = (\mu - r)/(\gamma\sigma^2)$ and

$$m^* = \frac{\rho - (1 - \gamma)[r + (\mu - r)^2/(2\gamma\sigma^2)]}{\gamma}.$$

At $(\rho, \gamma, \mu, r, \sigma) = (0.04, 2, 0.08, 0.02, 0.2)$ the targets are $(m^*, \pi^*) = (0.04125, 0.75)$. Wealth remains positive under this proportional policy. Its discounted terminal candidate expectation decays at rate $\rho - bg(m^*, \pi^*) = m^* > 0$, verifying its own transversality condition. The usual verification comparison is made over policies for which the stopped local martingale and discounted terminal inequalities pass to the limit.

The executed feature solver treats $q = \log A$, the consumption ratio, and the portfolio ratio as unknowns. It alternates the fixed-policy scalar evaluation root with an automatic-differentiation actor optimization, without feeding the analytical targets into the updates. Positivity is imposed by $A = e^q$. This is a learned homothetic feature model; the ratios alone would not be a neural-function benchmark on an unrestricted state space. The no-short regression at $\mu = 0.01$ instead has unconstrained share -0.125 and constrained share zero at every positive wealth. It is not a wealth-dependent switching problem.

S.4.2. *Epstein–Zin Driver and the Admissible Comparison Class*

Use $a = 1 - 1/\psi$, $b = 1 - \gamma$, with $a, b \neq 0$, $c > 0$ and $bV > 0$. The driver is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}.$$

The log limits must be taken separately when a denominator vanishes. For a homothetic candidate $V = AX^b/b$, dividing the stationary HJB by AX^b gives

$$\frac{\rho}{a} \{m^a A^{-a/b} - 1\} + g(m, \pi) = 0.$$

Thus a fixed proportional policy has a unique positive homothetic coefficient whenever $1 - ag/\rho > 0$. Its consumption first-order condition is $\rho m^{a-1} A^{-a/b} = 1$, and the portfolio first-order condition gives the same Merton share with risk aversion γ . Eliminating A yields

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2},$$

$$m^* = \psi\rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma\sigma^2} \right],$$

$$A^* = (\rho(m^*)^{a-1})^{b/a}.$$

For $(\rho, \gamma, \psi, \mu, r, \sigma) = (0.04, 5, 1.5, 0.08, 0.02, 0.2)$ this gives $m^* = 0.0455$ and $\pi^* = 0.3$. The normalized Hamiltonian is strictly concave in the consumption ratio and portfolio share on the specified positive domain. The selected ratios lie inside the implemented action bounds.

For precision about infinite horizon, the verification class consists of policies for which the policy-wise recursive equation exists on the branch $bY > 0$, its localized comparison argument is integrable, and the comparison-weighted terminal remainder vanishes. On compact subsets of $c > 0, bY > 0$ the driver is locally Lipschitz, but this does not imply a global Lipschitz constant on the entire half-line. One may first stop at compact localization sets, apply the mean-value linearization in the value argument, and then impose uniform integrability and the weighted terminal limit to remove localization. With $\Gamma_t = \exp(\int_0^t f_V(c_s, \zeta_s) ds)$ for the intermediate value ζ_s , the required remainder is of the form $\mathbb{E}[\Gamma_T |V(X_T) - Y_T^p|] \rightarrow 0$. The candidate HJB inequality and this comparison yield $Y_0^p \leq V(X_0)$, with equality for the maximizing policy.

For the selected homothetic policy itself, $f_V = -0.1115$, while the growth exponent of $\mathbb{E}X_t^b$ is $bg = 0.066$. The candidate comparison-weighted value therefore decays at rate $0.1115 - 0.066 = 0.0455 > 0$. The numerical record evaluates the fixed-policy equation and target errors, and the analytic calculation verifies this decay. This specifies a genuine stationary recursive benchmark with an explicit admissible verification class; it does not assert that all bounded controls satisfy the terminal condition automatically.

The sign-enforcing architecture is actually executed: $V = e^q X^b / b$, hence $bV = e^q X^b > 0$. A separate finite-horizon architecture test uses

$$V_\xi(t, X) = \frac{1}{b} \exp\{\log[bG(X)] + (T - t)N_\xi(t, X)\}$$

for $bG > 0$. It verifies the terminal identity and sign on test points. It is a domain test, not evidence that an additional finite-horizon recursive problem was trained. These distinctions follow the policy-wise recursive-utility formulation of [Duffie and Epstein \(1992\)](#) rather than replacing recursion by a pointwise evaluation at $c = 1, V = -1$.

S.4.3. A Structured Stochastic Dynamic Control Check

For $d \in \{4, 8, 16\}$, the additional six-period LQR problem uses

$$A = 0.82I + 0.07(J + J^\top), \quad B = 0.2I, \quad Q = I + 0.31\mathbf{1}^\top/d, \quad R = 0.15I,$$

and Gaussian shock loading $\Sigma = 0.08(I + 0.31\mathbf{1}^\top/d)$, with terminal matrix Q . Here J has ones on its upper adjacent diagonal. The linear neural actor and quadratic feature critic are trained by policy evaluation and improvement. The independent reference uses the finite-horizon Riccati recursion, including the noise contribution to the value constant. A second recursion evaluates the learned fixed policy rather than using its fitted critic as a performance score.

The experiment has cross-state dynamics and non-diagonal stochastic covariance but no binding control constraints. It is retained as a structured dynamic regression test. The main shared-budget resource experiment supplies the separately executed constrained case; neither is relabeled as an adaptive sparse-grid comparison.

TABLE S.2
STRUCTURED STOCHASTIC LQR REGRESSION

d	Actor coeff. error	Cost excess max.	Neural sec.	Riccati sec.
4	4.25e-06	3.03e-11	0.0571	0.00037
8	2.25e-06	2.14e-11	0.0724	0.00028
16	4.30e-06	3.84e-11	0.1330	0.00033

Six periods, linear actors and quadratic feature critics, independent Riccati reference, and separate frozen-policy evaluation. Cost excess is normalized per dimension in the implementation. This structured unconstrained regression is not the main binding-resource experiment and supplies no generic neural speed advantage.

S.5. TEMPORAL SELVES AND DYNAMIC COMPETITION

S.5.1. *Sophisticated Consumption–Savings*

Let one period have length Δ , $\delta = e^{-\rho\Delta}$, and terminal payoff $G(w) = \log w$. The acting self chooses $c \in (0, w)$ to maximize $\Delta \log c + \beta \delta V_{n+1}(w - c)$, whereas ordinary continuation evaluates the chosen policy as

$$V_n(w) = \Delta \log c_n(w) + \delta V_{n+1}(w - c_n(w)).$$

If $V_{n+1} = A_{n+1} \log w + B_{n+1}$, strict concavity gives

$$\begin{aligned} z_n &:= \frac{c_n(w)}{w} = \frac{\Delta}{\Delta + \beta \delta A_{n+1}}, \\ A_n &= \Delta + \delta A_{n+1}, \\ B_n &= \Delta \log z_n + \delta A_{n+1} \log(1 - z_n) + \delta B_{n+1}. \end{aligned}$$

Starting with $A_N = 1, B_N = 0$ yields an exact sophisticated equilibrium. The coefficient A_n is independent of β , but the policy and the constant are not. The reference test uses $\Delta = 1$, four consumption dates, $\rho = 0.04$, and $\beta \in \{0.7, 1\}$. At four wealth levels it separately solves the bounded one-shot consumption problem and checks the evaluation identity with the frozen continuation coefficients.

For two consumption dates, $\delta = 1$, the correct first-date ratio is $1/(1 + 2\beta)$. Reusing the acting self's maximand as its stored continuation would instead yield $1/(1 + \beta + \beta^2)$. At $\beta = 0.7$ the difference is 0.4166667 versus 0.4566210. The $\beta = 1$ check alone cannot reveal this error. These checks implement the written temporal game rather than an ordinary geometric-discount control problem. They do not identify the finite-state reference as a trained neural solution.

S.5.2. *A Stochastic Capacity Duopoly*

There are four decision dates. Demand is $D \in \{1, 1.3\}$ with probability 0.8 of staying at its current value. Each firm's installed capacity $K_i \in \{0, 1\}$ determines current output $q_i = K_i/3$. Current operating profit is $q_i(D - q_i - q_j)$. Firm i chooses a maintenance or renewal intensity $a_i \in [0, 1]$ at cost $0.4a_i^2/2$. Independently across firms conditional on actions, next-period capacity is one with probability a_i and zero otherwise. The discount factor is 0.95 and terminal value is $0.05K_i$. There are eight Markov states and four dates, so complete state enumeration is possible.

For fixed rival intensity, the continuation value is affine in a firm's own intensity. Its best response is therefore the projection onto $[0, 1]$ of its discounted marginal continuation gain divided by 0.4. The joint map is affine before projection, with slope determined by the cross difference of the four next-capacity continuation values. The program checks the product of the two own best-response slopes, iterates from multiple initial profiles, and compares their limits. This finite example has a contraction bound below one. It does not establish uniqueness in arbitrary dynamic games.

An independent backward best-response recursion freezes the rival's entire state-contingent policy at every subsequent date and allows the tested firm to deviate at all dates. Its difference from the profile value is dynamic exploitability. This is stronger than checking a single current Hamiltonian against an inconsistent continuation. The report records the maximum over all dates, states, and players, with a numerical tolerance of 10^{-10} . A separate automatic-differentiation test confirms that a player's loss does not differentiate the rival's parameters.

The static special case $q_i(1 - q_i - q_j)$ remains a useful negative control. Symmetric Nash output is $1/3$, whereas joint-profit output is $1/4$. At the latter profile, changing one firm's quantity to its own best response is profitable. A symmetric or jointly optimal actor graph is therefore not a substitute for a Nash calculation. The dynamic capacity model is a finite-state equilibrium computation; no competitive limit or general neural-game convergence is inferred from it.

S.5.3. Market-Size Normalization and a Conditional Small-Player Bound

To retain the economically relevant many-firm extension without conflating different economies, consider per-firm normalized output y_i and actual output $q_i = y_i/N$, with inverse demand $P = a - b\bar{y}$, $\bar{y} = N^{-1} \sum_i y_i$. Multiplying each firm's monetary payoff by N gives normalized flow $y_i P - C(y_i)$ under consistently scaled costs. Its direct strategic price derivative is $-by_i/N$. Investment, depreciation, and state transitions must be scaled and retained as well; this current-flow derivative by itself does not establish convergence of dynamic equilibria.

A precise conditional implication is available. Suppose that, against a fixed rival profile, the finite- N player's transition operators and a price-taking comparison operator are monotone and γ_n -Lipschitz, have the same terminal payoff, and differ uniformly by at most η_n/N on all policies and continuation values used in the comparison. Then a policy optimal for the price-taking problem loses at most

$$\frac{2}{N} \sum_{j=n}^{N-1} \Gamma_{n,j} \eta_j$$

relative to the finite- N best response. To prove it, backward iteration bounds the difference of the two optimal values by $N^{-1} \sum_j \Gamma_{n,j} \eta_j$, and bounds the two evaluations of that fixed policy by the same amount. Their sum gives the result. Uniform Lipschitz control of aggregate-state effects is one route to the operator hypothesis, but it must be checked in a specified model. This proposition concerns a unilateral deviation bound, not selection or convergence of a sequence of equilibria. It is theoretical extension material and is not recorded as an executed large- N experiment.

S.6. REPLICATION AND ADDITIONAL RESULTS

This section retains the foundational neural experiments and their full checkpoint provenance. The fifth-revision additions and independent rechecks are described in S.7–S.8.

S.6.1. *Source and Evidence Identities*

The authoritative R4 source record consists of `ECTA_R4.tex`, `SUPP_R4.tex`, their explicitly included sections, and `replication/r4/`. The base review is pinned at commit `79a7d84`. The replication manifest records the reachable source commit, full SHA-256 digests of the source files, package versions, and output identities. Source files are committed before their authoritative execution. A later result commit can add generated outputs without changing that source anchor.

Execution status, evidence class, and numerical-tolerance status are separate fields. A completed program is not automatically a solved control problem. The coupled resource runs have a specified held-out cost-loss target and independent reference gaps. The NDU comparison reports discrepancies and does not assign a passed uniform PDE certificate. Algebraic checks, domain tests, finite-state equilibria, and structured feature solvers each have their own evidence class. An unavailable continuous-time error bound or reflection regulator is null with an explanation, not a numerical zero.

Scientific digests exclude elapsed time and hardware measurements. File digests still identify complete files, including timing records. Floating-point algorithms, package versions, and BLAS implementations can change last digits across platforms. The source and saved weights permit direct policy replay independently of retraining. Timings include neural fitting and full evaluation for the stated initial-state batch; they do not imply a universal hardware comparison or unmeasured amortization advantage.

S.6.2. *Architectures and Optimization*

For the stopped preference model the safeguarded neural run uses separate two-hidden-layer tanh actors of width 32 and critics of width 64 at each of eight decision dates. The actor has three bounded output coordinates. The critic writes the value as G plus remaining time times a product vanishing on all spatial faces times its network output. The training sample contains 2,048 interior states. L-BFGS caps are 160 actor and 1,000 critic iterations with strong-Wolfe line search; actual function evaluations and iteration counts are recorded when produced by the optimizer. The implementation never trains on the independent grid reference’s values or policies.

The unsafeguarded pilot uses width-32 critics and 1,024 training states with a smaller critic budget. It is retained as a development baseline, not used as a matched causal ablation. The resource experiment supplies a matched ablation: the same fitted actor weights are evaluated with and without pointwise improvement at deployment. It holds training, initial states, shocks, and model primitives fixed.

For resources the convex critic is

$$q(x) + \ell^\top(x - \mathbf{1}) + c_0 + \sum_{j=1}^{6d+102} \alpha_j [\max\{w_j^\top(x - \mathbf{1}) + b_j, 0\}]^2, \quad \alpha_j \geq 0.$$

Directions consist of signed coordinate axes, signed aggregate directions, and 32 seeded Gaussian directions, each at offsets $-0.5, 0, 0.5$. The coefficients are fitted by bounded

TABLE S.3
ALL RESOURCE RUNS AND THE MATCHED DEPLOYMENT ABLATION

d	Seed	NBO upper loss	Proposal only	Reference gap	Binding at 0
4	101	6.45e-04	1.19e-02	8.32e-08	0.812
4	202	4.88e-04	8.14e-03	7.40e-08	0.938
4	303	5.23e-04	6.31e-03	7.02e-08	0.938
8	101	4.29e-04	9.54e-02	9.79e-08	0.938
8	202	4.61e-04	3.28e-02	9.97e-08	1.000
8	303	2.03e-04	3.89e-02	9.75e-08	1.000
16	101	5.41e-04	6.73e-02	9.54e-08	1.000
16	202	5.67e-04	6.81e-02	9.63e-08	1.000
16	303	5.24e-04	6.65e-02	6.02e-08	1.000

Each row uses the stored 32 initial states and every one of the 64 terminal shock branches. The reference gap is the complete scenario-tree linear minimization gap at the tighter audit tolerance. Binding is the fraction of initial actions exhausting the common budget. All declared dimension–seed combinations are included.

least squares with a 10^{-5} ridge term. The actor has one width-32 tanh layer and a slack-softmax output enforcing the common budget. Its 160-iteration L-BFGS fit distills feasible greedy actions. The convex pointwise improvement uses an explicit gradient-Lipschitz upper bound and checks its linear minimization gap rather than equating a stationary point with a maximum. All three seeds, 101, 202, and 303, are included at each dimension.

The independent scenario tree contains one root action, four next-date actions, and sixteen last-date actions for each tested initial state. Its objective sums all state and control costs and the 64 terminal costs with their probabilities. An independent backward adjoint computes the gradient. Reference optimization stops at a common-accuracy gap of 10^{-3} for timing and is separately continued to 10^{-7} for policy-cost auditing. Nonanticipativity is enforced by one action per information node, not one freely chosen action per complete shock path.

S.6.3. *Reproduction Order*

The repository README gives executable commands for the following sequence: generator, temporal, trace, utility, and game tests; stochastic grid refinements and re-solved adjustment-cost panels; the retained unsafeguarded neural pilot; all three safeguarded NDU runs; nine resource runs; independent frozen-policy and occupation analysis; computational-graph and checkpoint-replay tests; table generation; source/output manifest generation; and LaTeX builds. The table generator reads the deposited JSON records, so manually entered historical arrays cannot silently become current numerical evidence.

The source snapshot also retains the previous manuscripts, supplements, reports, reviewer diagnostics, and arrays unchanged. The preservation map locates their substantive topics in the new manuscript or this supplement. Retention is not endorsement of earlier incorrect claims. The development log records the standard-library module-name collision discovered during implementation, the corrected execution, and the budget difference of the pilot. Failed commands are not relabeled as successful scientific experiments.

S.6.4. *What Each Numerical Error Means*

For NDU, a reference value difference compares two specified finite approximations or evaluates a neural policy against a specified finite reference. The actor discrepancy reports consumption, preference adjustment, and portfolio separately because they have different units. A feasible Bellman gain is measured in continuation-value units and uses an independently evaluated frozen continuation. It is a lower bound on the maximum profitable one-step deviation over a richer action set, not an upper bound on all deviations.

For resources, the full stochastic tree gives the evaluated policy cost. The reference tangent gap bounds its unknown optimum on that same tree, resulting in the interval in S.1. The held-out initial states are explicitly stored, so the bound is not extrapolated to all states. The Merton and Epstein–Zin policy errors compare learned homothetic coefficients with analytical solutions, while LQR errors compare structured learned policies with an independent Riccati solution. Temporal and game exploitability use separate best-response calculations against the actual continuation being evaluated. Trace standard errors concern a repeated-batch scalar loss and are not policy errors.

S.7. STOPPED CONTRACTS AND VALUE-ENCLOSED POLICY REUSE

S.7.1. *Discounted Duration and Terminal Exposure*

For any realized stopping time $\tau \leq T$, direct integration gives

$$\rho \int_t^\tau e^{-\rho(v-t)} dv + e^{-\rho(\tau-t)} = 1.$$

This equality holds before conditioning and before optimizing. In the positive finite model, each branch satisfies the same identity with its actual live duration αh . Thus the independently evaluated exposure recursion is

$$Z_n^p(s) = z_n(s, p_n(s)) + K_n^{p_n} Z_{n+1}^p(s), \quad Z_N^p = 1,$$

where z_n is the discounted probability weight of settlement during the current branch and K_n^p contains only live discounted interpolation weights. Duration satisfies $A_n^p = a_n^p + K_n^p A_{n+1}^p$, $A_N^p = 0$. The row identity $\rho a_n^p + z_n^p + K_n^p \mathbf{1} = 1$ proves $\rho A_n^p + Z_n^p = 1$ by induction. This includes immediate settlement for an initial boundary node.

The program does not define Z as $1 - \rho A$ and then test the resulting tautology. It evaluates Z with its own killed-payment recursion and checks the identity for randomly selected contracts and stored-policy switches. It then compares the original four-feature payoff $B + mA - kC + \ell Z$ with the reduced payoff $\ell + B + (m - \rho\ell)A - kC$. The largest discrepancy and the source hash are recorded. For state paths generated by an optimal-policy selection, all changes $(m, \ell) \mapsto (m + \rho h, \ell + h)$ preserve the set of feasible optimal laws; this is the precise sense in which choices alone do not identify the two levels separately.

The monotonicity result concerns a common transition and feasible-policy class. Changing the shock correlation, the control bounds, or the stopping mechanism does not satisfy that condition. Nor does a different parameter-specific action menu necessarily preserve convexity of the optimized finite value in the original parameters. We therefore freeze the complete union menu over the contract interval before building the library.

S.7.2. Certified Anchors and Approximate Policy Features

Let a feasible anchor policy p_i have certified loss $c_n^i(s)$ at its anchor q_i . Exact evaluation gives the valid upper bound $\bar{V}_n^i = J_n^i(q_i) + c_n^i$. If evaluation instead supplies an interval $[J_n^i, \bar{J}_n^i]$, the valid anchor upper bound is $\bar{J}_n^i(q_i) + c_n^i$. The empirical maximum of sampled actor residuals cannot be substituted for either ingredient.

For clarity, an error version of switching can be stated without conflating approximate policy features with exact evaluation. Suppose $\|\hat{J}_n^i - J_n^i\|_\infty \leq e_n$ uniformly over stored policies at the query, and choose the policy index maximizing $\hat{J}_n^i$. Let $L_n = \max_i J_n^i$ and $D_n = \|(L_n - J_n^{\text{sw}})_+\|_\infty$. If continuation kernels have row sums at most γ_n , then

$$D_n \leq 2e_n + \gamma_n D_{n+1}, \quad D_n \leq 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j \quad (\text{S.1})$$

with common exact terminal evaluation. To prove the first inequality, the selected policy has exact value at least $L_n - 2e_n$. The next switching value is at least the selected continuation value minus D_{n+1} ; positivity gives the result. Hence

$$V_n^* - J_n^{\text{sw}} \leq U_n - L_n + 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j.$$

The exact-feature theorem in the main paper is the $e_n = 0$ case. The executed finite recursion uses this case up to recorded floating-point errors; it does not assign zero error to the neural critic. This distinction allows neural approximation to provide proposals while the certificate uses separately evaluated policies.

S.7.3. A Finite Certificate for an Entire Interval

For adjacent anchors $a < b$, two feasible endpoint-policy lines are $b_1 + da_1$ and $b_2 + da_2$, independently at every date and state. With an affine upper chord $u_0 + du_1$, define

$$g(d) = u_0 + du_1 - \max\{b_1 + da_1, b_2 + da_2\}.$$

On either side of the intersection of the two lines, g is affine. Its maximum on $[a, b]$ must therefore occur at a , at b , or at their intersection if present. Parallel lines require only the endpoints. Taking the maximum over the finite set of dates and states gives a uniform bound without discretizing the contract interval. Enlarging the policy library can only improve the maximum of feasible policy values; using just the endpoint pair to certify an interval is conservative and reduces certification cost.

The adaptive construction begins with contracts zero and one. At every iteration it computes the complete bound, selects a worst interval, and solves a new anchor at a parameter attaining that interval's largest gap. Each new policy is independently evaluated. The complete anchor sequence, worst gaps, and parameter locations are saved, including the stopping criterion. The implementation has a maximum-anchor safety limit that raises an error rather than publishing an uncertified library when the tolerance is not reached.

All computations use double precision. The pairwise-intersection calculation treats duration-slope differences below 10^{-14} as parallel. The omitted variation of a difference of such lines on the unit contract interval is at most 10^{-14} ; the verification checks

use a larger 10^{-9} slack. The published 10^{-3} threshold has substantial slack relative to both quantities. This is a numerical certificate based on explicit finite inequalities, not an interval-arithmetic proof of every floating-point operation. Independent policy recursions, kernel comparisons, and direct query solves test the implementation of the inequalities.

S.7.4. *Whole-Interval Action Separation*

At a fixed date and state, choose one feasible first action a and one library continuation p_i . Both its reward and its continuation value are affine in the contract. For each competing first action b , the upper-chord continuation gives another affine function. If their difference is positive at both interval endpoints, it is positive at every convex combination of the endpoints. Taking a supremum over all competing actions then proves the main action-separation proposition.

The implementation maximizes the minimum of the two endpoint margins over candidate pairs (a, p_i) . This ordering of maximization and minimization is essential. Taking the minimum of two independently maximized endpoint margins would permit a different favorable witness at each endpoint and would not establish a single affine lower bound on the interval. Saved output includes the selected library index, the first action, both sign margins, and the reported status of each interval. The displayed negative-region endpoint is rounded downward and the positive-region endpoint upward; neither rounded interval extends beyond its certificate.

S.7.5. *Relation to Existing Policy-Transfer Results*

The feature decomposition and the policy-improvement logic build on [Barreto et al. \(2017\)](#), [Chang \(2021\)](#). The anchor upper bound is the convex-combination case of [Nemecsek and Parr \(2021\)](#), whose policy-cache method already addresses whether an existing policy is sufficiently accurate for a new reward function. Our construction uses that bound on the economically reduced stopped-contract parameter, checks every interval breakpoint and every finite state-date node, and applies a separate interval-wide action-separation inequality. The independently verified finite-economy results and the stopping-contract interpretation are the substantive objects reported here. We do not claim that the underlying supremum-of-affine-functions argument, policy caching, or actor-critic separation was first introduced in this paper.

S.8. ADDITIONAL COUNTERFACTUAL AND EQUILIBRIUM COMPUTATIONS

S.8.1. *Joint Refinement and Contract Sensitivity*

The complete refinement panel is reported in Table [S.4](#). Unlike refining time on a fixed interpolation grid, these calculations change state spacing, time spacing, and the action mesh together. They retain the consumption and preference controls as active optimization coordinates. The coarse, intermediate, and fine initial values at $d = 0$ are different; the portfolio at the midpoint is also sensitive to the first refinement. The endpoint contract comparison is more stable than the estimated switching location. All entries remain evaluations of declared finite economies rather than upper and lower bounds on the stopped diffusion.

The correlation panel in the machine-readable output independently resolves the intermediate economy at correlations -0.25 , 0 , and 0.25 . For each, the initial share

TABLE S.4
JOINT TIME, STATE, AND ACTION REFINEMENTS

State grid	Dates	Action grid	d	Value	Duration	π_0
17×25	4	$5 \times 5 \times 7$	0.0	-1.140548	0.884705	-0.50
17×25	4	$5 \times 5 \times 7$	0.5	-0.698148	0.885220	-0.50
17×25	4	$5 \times 5 \times 7$	1.0	-0.253215	0.891936	0.80
33×49	8	$7 \times 7 \times 11$	0.0	-1.136633	0.887830	-0.50
33×49	8	$7 \times 7 \times 11$	0.5	-0.692395	0.890237	0.80
33×49	8	$7 \times 7 \times 11$	1.0	-0.246813	0.894453	0.80
49×73	16	$9 \times 9 \times 13$	0.0	-1.133946	0.888358	-0.50
49×73	16	$9 \times 9 \times 13$	0.5	-0.689493	0.890188	0.80
49×73	16	$9 \times 9 \times 13$	1.0	-0.243752	0.894848	0.80

All controls are reoptimized at every resolution. These refinements use the displayed meshes without the extra frozen neural proposals. The endpoint reversal survives, whereas the midpoint changes sign between the first and second resolution. This table measures sensitivity of finite economies, not a certified bound for the continuous stopped diffusion.

TABLE S.5
ADJUSTMENT QUANTITY AND COEFFICIENT-WEIGHTED EXPENDITURE

k	C^{p_k}	kC^{p_k}	Value
0.5	0.014614	0.007307	-1.117524
2.0	0.009888	0.019777	-1.135835
8.0	0.001432	0.011455	-1.160291

The complete finite problem is solved separately at each coefficient on a 25×37 grid with eight dates and $9 \times 9 \times 13$ actions. The unweighted quantity is decreasing; the coefficient-weighted charge is not. Stopping is included in both quantities.

changes from -0.5 at $d = 0$ to 0.8 at $d = 1$. We do not transfer the feature bank across these changes in transition law. Duration, values, and all controls are stored for each separate solve. The normalization comparison, by contrast, changes only the reward coefficients and retains the same transition and policy class.

Table S.5 distinguishes discounted adjustment effort from its coefficient-weighted charge. The same decomposition is evaluated at the original cost-panel grid and with the original complete control vector. The theorem applies to the first quantity; the middle coefficient produces the largest weighted charge in this panel.

S.8.2. Matched Actor-Free and Quadratic Comparisons

The nine actor-free comparisons reload the deposited nonlinear critic weights and the same initial states. Both actor-start and zero-start algorithms use the same projected convex improvement method, gradient, Lipschitz step bound, and stopping tolerance. The full policy is then evaluated through all 64 terminal shock paths. Saved output separates actor initialization, optimization time, iterations at each date, complete life-time cost, and the reference gap. The original proposal-only ablation remains in S.6; removing the safeguard and removing the actor are different experiments.

To isolate training costs, critic-only backward fits use the same training-state draws and feature construction as the original nonlinear critic. Actor distillation is timed

TABLE S.6
ISOLATED ACTOR COSTS AND INITIALIZATION SAVINGS

d	Seed	Actor train	Actor query	Zero query	Projected break-even
4	101	1.483	0.0414	0.0424	48512
4	202	0.741	0.0432	0.0451	12544
4	303	0.742	0.0436	0.0429	None
8	101	0.808	0.0607	0.0626	13568
8	202	0.797	0.0637	0.0713	3392
8	303	0.794	0.0489	0.0512	11488
16	101	0.878	0.1004	0.1035	8864
16	202	0.879	0.1174	0.1326	1856
16	303	0.875	0.1051	0.1079	9888

Seconds per 32-state complete-tree query batch. Actor training is the distillation fit alone; common target construction and critic training are not charged twice. The last column is $32 \lceil T_{\text{actor}} / (t_{\text{zero}} - t_{\text{actor}}) \rceil$ when the measured saving is positive. It is a linear fixed-batch projection, not an observed crossover. “None” means that actor initialization was not faster in that timing comparison. Optimizer iteration counts and separated initialization times are saved in the machine-readable results.

separately against the same improved-action targets. Table S.6 reports the complete nine-case timing panel. A positive measured online saving is converted into a clearly labeled fixed-batch break-even projection. The separately executed fresh-state workloads in the main paper, not this projection, support the observed amortization comparison.

The quadratic comparator minimizes squared fitting error over

$$q(x) + b + \ell^T z + z^T H z, \quad z = x - \mathbf{1}, \quad H \succeq 0.$$

The regression is convex in (b, ℓ, H) . Its gradient is computed from the full quadratic design matrix. Projected accelerated gradient leaves (b, ℓ) unrestricted and projects the symmetric matrix onto the positive-semidefinite cone by clipping negative eigenvalues. A small ridge term stabilizes the design. The reported projected-gradient norm is below the declared 10^{-8} stopping threshold in the executed fits. At deployment the analytic gradient is $\nabla q(x) + \ell + 2Hz$, and its Hessian bound is the largest eigenvalue of $\nabla^2 q + 2H$. The common investment projection and full scenario-tree cost calculation are unchanged. This comparison tests a genuinely coupled convex quadratic, rather than a separable or unoptimized baseline.

The fresh-state workload charges only for the computations used by each competing method. The reusable policy’s cost is critic training plus zero-start online improvement and full policy evaluation. The specialized solver’s cost is a complete nonanticipative reference optimization at tolerance 10^{-3} . A reference solved more tightly to gap below 10^{-7} supplies the accuracy upper bounds for both; it is not charged asymmetrically as part of one solver’s online cost. Hardware, dependencies, seeds, and individual times accompany the outputs. A three-period, rank-two-shock experiment cannot establish dimension-independent performance, even when its measured query reuse is useful.

S.8.3. Persistent Capacity and All-Date Dynamic Deviations

Let $b_i = 1 - sK_i$, so that next-period capacity probability is $sK_i + b_i a_i$. Write $W_i(k'_1, k'_2)$ for player i ’s continuation value after averaging over next demand. For player one the

best response is

$$a_1 = \text{clip} \left(\frac{\delta b_1}{\kappa} \{W_1(1,0) - W_1(0,0) + (sK_2 + b_2a_2)\Delta W_1\}, 0, 1 \right), \quad (\text{S.2})$$

where $\Delta W_1 = W_1(1,1) - W_1(1,0) - W_1(0,1) + W_1(0,0)$. The corresponding formula holds for player two. The action criterion is strictly concave because the continuation is affine in the player’s own probability and the action cost has curvature $\kappa > 0$. Enumerating whether each clipped best response is at zero, interior, or one gives nine active-set combinations. Feasible candidates are checked against both original best-response equations.

The product of the two affine slopes bounds the Lipschitz constant of the composed best response; clipping is nonexpansive. Its recorded maximum below one therefore provides a unique one-period fixed point for every continuation game. Backward induction supplies a finite-horizon Markov-perfect profile. This argument does not assume current-capacity independence and does not rely on convergence from a single initialization.

For independent dynamic verification, let $J_{i,n}$ be the value of the frozen profile and define the full unilateral best-response value

$$B_{i,n}(s) = \max_{a_i} \{r_i(s, a_i, p_{-i,n}(s)) + \delta \mathbb{E}[B_{i,n+1}(s') \mid s, a_i, p_{-i,n}(s)]\}.$$

At *every* backward date, the code stores $B_{i,n}(s) - J_{i,n}(s)$ for both players and all eight states. It reports the largest positive gain, the signed maximum and minimum, and the attaining date, player, capacities, and demand. The historical code had placed the comparison after the date loop, which verified only date zero. The corrected scope is executed for both the renewal regression and the persistent economy.

As a negative control, the final-period investment of player one at capacities (1,1) and low demand is replaced by one. The independent routine recomputes the frozen-profile value and all best responses. It detects a gain exceeding 0.1, at a recorded late-date node. This test would not be passed by a routine that merely reports convergence of the equilibrium solver. The reset case $s = 0$, four dates, reproduces its early closed-form actions and terminal investment to numerical precision. Both examples remain exact finite-state references, not neural-equilibrium training results.

S.8.4. Executable Files and Preservation

The fifth-revision package is in `replication/r5`. It imports the unchanged fourth-revision transition, resource primitives, and saved neural checkpoints. The contract computations are in `contracts.py` and `economic_robustness.py`. The remaining experiments are in `resource_ablation.py` and `persistent_game.py`. The table generator reads executed JSON; the validation script replays policy features, normalization identities, kernel comparisons, and certificate conditions. The build records input and output hashes and preserves the exact review-base commit.

Earlier papers, theorem proofs, raw negative controls, referee reports, and neural checkpoints remain in their original paths. The new manuscript carries the original mathematical and economic extensions forward and adds the contract-reuse and persistent-capacity material. The accompanying response identifies where each issue in the fourth-revision report is treated. Build completion and numerical verification do not assert journal submission or editorial approval.

S.9. TRANSITION-LAW CERTIFICATES AND MECHANISM REPLICATION

S.9.1. Coefficient Construction and Interval Bounds

For a fixed policy, pre-draw the h remaining iid regime labels. Conditional on their count j , labels are exchangeable, the first label is 1 with probability j/h , and the remaining count is $j - 1$ or j . Applying the same count-blind policy to these conditional paths gives the main coefficient recursion. Averaging over the binomial count distribution proves the polynomial representation. This is an evaluation identity, not a policy allowed to condition its action on the count.

For the separate upper calculation the controller is allowed to observe the remaining count, and past labels are revealed after their transitions. Every original policy can ignore this extra information. Backward maximization of the weighted common-action criterion therefore gives an upper coefficient vector. Averaging over counts preserves the upper inequality. Optimizing the regime-0 and regime-1 actions separately *before* taking their weighted mean would instead disclose the current label and define a different relaxation. The implementation takes the maximum after the mean.

If $h \geq 2$, the degree- h Bernstein coefficients of the corrected upper value on a local interval are

$$U_{n,j} = \left(1 - \frac{j}{h}\right) v_n^a + \frac{j}{h} v_n^b + \frac{j(h-j)}{h(h-1)} M_n, \quad j = 0, \dots, h.$$

At $h = 1$ the correction is zero because the common terminal payoff has zero endpoint difference; at $h = 0$ the only coefficient is g . Degree elevation thus represents the quadratic upper function exactly. De Casteljau subdivision restricts both upper and lower coefficients to the same closed cell before taking their coefficient differences. The bound is conservative because $\sup_t \min_p f_p(t) \leq \min_p \sup_t f_p(t)$. The minimum is taken separately for each state and date, not before the statewise maximum.

The stopping condition uses the maximum of these cell bounds, not the largest loss observed at replay parameters. The replay additionally checks the whole value vector, dominance by the upper function, and improvement from switching relative to the best stored fixed policy. The script stores each anchor policy, its independently evaluated polynomial coefficients, each interval's upper coefficients, cell bounds, and validation locations. The count-information and corrected-chord outputs remain separate so that a loose information bound cannot be substituted for the stopping certificate.

S.9.2. A Reusable QP with the Original Information Structure

At a tree node, write its state as $x_h = M_h x + S_h z + b_h$. Let $Q = I/d + 0.211^\top/d^2$ and let p_h be its probability. Expansion of all stage and terminal costs gives

$$H = 2 \sum_h p_h S_h^\top Q S_h + 2 \text{diag}_h(0.1 p_h I/d), \quad F = 2 \sum_h p_h S_h^\top Q M_h, \quad f_0 = 2 \sum_h p_h S_h^\top Q (b_h - \mathbf{1}).$$

The control diagonal includes decision nodes only. The sums over state costs include terminal nodes. Let D repeat each decision node's probability across its d controls. A Lipschitz constant for the transformed gradient is the largest eigenvalue of $D^{-1/2} H D^{-1/2}$. Each accelerated step is a nodewise simplex projection of $z - D^{-1} \nabla C_x(z)/L$. Every ten iterations we check the complete convex tangent gap; the final gap is independently recalculated by the original tree-gradient routine. The reported query time includes that

evaluation. The structural setup and the learned training times are both included in their respective totals; the tighter deposited reference enters only the accuracy comparison.

The assembly is a new source implementation of the structural comparison described in the fifth-round report, not an invention of parametric quadratic programming. All cold-start workloads use the same saved initial states. No array of known query solutions initializes a new query. Complete costs, gaps, timing samples, feasibility errors, and gradient comparisons are retained in the execution record.

S.9.3. Risk-Class Evaluation and Scope

For a given (d, k) , first solve the original finite problem backward. At the initial state, optimize the current action separately over positive and nonpositive portfolio positions using the same optimal continuation. Evaluate each resulting complete policy independently to obtain (B, A, C) . This computes first-risk-class values, not policies constrained to keep the same sign forever. Repeat the entire backward solve with $\theta = 0$ at every state and date for the no-deliberate-adjustment model. Preference volatility remains present. The two risk-specific option values are nonnegative by set inclusion, and their difference is checked against the independently formed difference of risk advantages.

Bisection starts only when the risk advantage has opposite endpoint signs. The output records both signs, the bracket, the active class features, and the ratio of their effort and duration differences. It does not assign a global uniqueness flag. A duration-gap lower bound valid throughout an interval would justify the main uniqueness result; the present numerical brackets do not assert that additional hypothesis. The portfolio-sign exclusions in the inherited reward experiment therefore retain their original independent status.

S.9.4. Evidence and Reproduction

The current source record is `ECTA_R6.tex`, `SUPP_R6.tex`, the sixth-revision section directory, and `replication/r6/`. Sections S.1–S.8 preserve earlier substantive proofs, specifications, and evidence with their historical identities. The new manifest pins the reachable source commit and hashes the executable sources, immutable input checkpoints, generated records, and tables. The public execution log distinguishes source preparation, experiments, validation, and document build.

The support comparison uses `bank_comparison.py`. Transition calculations use `transport.py` and `kernel_certificate.py`. The remaining programs are `mechanism.py`, `structural_qp.py`, and the independent small-model checks in `unit_tests.py`. Numerical libraries are single-threaded for timing. Two developmental transition-certificate runs were stopped to eliminate repeated interval calculations and then to replace array multiplication by an equivalent CSR multiplication. The final runs use the cached intervals and verified CSR rows. This is an execution optimization, not a change in the state, action, transition, or welfare target. The all-state scalar-OLS comparison retains the inherited common oracle implementation for both objectives.

The revised source does not reclassify the historical neural fits, exact grid repairs, or finite games. In particular, removal of neural proposals still meets the original reward-bank welfare target, and the reusable QP still supplies the relevant stronger resource comparator. A completed build or a passing finite-model validation does not establish journal acceptance, broad neural superiority, or a diffusion approximation theorem.

S.10. COMPARATIVE UPPER ORACLES AND TOTAL POLICY-BANK ERROR

This section supplies the additional proofs for the matched comparison and the finite-refinement result. All statements concern the same finite state/action target and a scalar affine mixture of positive substochastic operators. Approximation of a diffusion is a separate question.

S.10.1. *Localized Count Dominance*

Fix an interval $[a, b]$, let $h = N - n$, and suppress its superscript. Write $C_{n,j} = (1 - j/h)v_n^a + (j/h)v_n^b$ and $f_{h,j} = j(h - j)/(h(h - 1))$ for $h \geq 2$. The corrected upper coefficients are $U_{n,j} = C_{n,j} + f_{h,j}M_n$. At $h = 0$, both constructions equal g . At $h = 1$, the correction is zero because both endpoint terminal functions are g , and the pure endpoint Bellman backups are no greater than their endpoint supersolutions.

Suppose $W_{n+1,j} \leq U_{n+1,j}$. Positivity permits substitution of U for W in the count recursion. For an interior count, expanding the part containing C gives

$$\begin{aligned} (1 - j/h)T_{n,a}^u v_{n+1}^a + (j/h)T_{n,b}^u v_{n+1}^b \\ + f_{h,j}(K_{n,a}^u - K_{n,b}^u)(v_{n+1}^b - v_{n+1}^a) \leq C_{n,j} + f_{h,j}D_n^u. \end{aligned}$$

This identity includes the reward terms, whose weights sum to one. For $h > 2$, the propagated correction is

$$f_{h,j} \left\{ \frac{h-j-1}{h-2} K_{n,a}^u M_{n+1} + \frac{j-1}{h-2} K_{n,b}^u M_{n+1} \right\}.$$

The two weights are nonnegative and sum to one. Keeping the current action u fixed, the entire expression is at most

$$C_{n,j} + f_{h,j}\{D_n^u + \max(K_{n,a}^u M_{n+1}, K_{n,b}^u M_{n+1})\} \leq C_{n,j} + f_{h,j}M_n.$$

For $h = 2$ the next correction is zero and the same conclusion follows from D_n^u . Counts zero and h use the endpoint supersolution inequalities directly. Maximization over the common action completes induction. This is the comparison identified in the R6 advisory report, including its same-action and short-horizon qualifications.

For the rectangular comparison, induction gives $W_{n+1,j} \leq B_{n+1}$. Each count backup is then a convex combination of two quantities no greater than $\max_{u,e} T_{n,e}^u B_{n+1} = B_n$, so $W_{n,j} \leq B_n$. Each original policy can ignore count information, proving that the count polynomial itself is an upper value. Positive polynomial evaluation and positive de Casteljau subdivision preserve coefficient dominance. Subtracting the same lower coefficients, taking coefficient maxima, policy minima, positive parts, and state/date maxima preserves the ordering of the certificates. These statements do not establish that count is faster.

S.10.2. *Operation Counts and Storage*

With exact endpoint values shared, count coefficients $j = 0, h$ need no additional Bellman solve. At remaining horizon $h \geq 2$, all h next-date coefficients require one multiplication by each base kernel. Neighboring results can be cached while looping through j . The total is $2 \sum_{h=2}^H h = H(H+1) - 2$ action-wide kernel applications. The

correction has one two-application nonzero layer at $h = 2$ and four applications at each later layer, totaling $4H - 6$ for $H \geq 2$; it is zero for $H = 1$. The rectangular recursion uses two applications per date. Reward combinations and maxima are additional vector operations. The production code's counts are checked by an instrumented dense backend at horizons 1, 2, 3, 4, 6, 8, 12, 16.

For S states, the correction stores $H - 1$ nonzero state vectors; known zero vectors are not retained in its compact representation. The count method has $\sum_{h=1}^H (h - 1) = H(H - 1)/2$ interior coefficient vectors. Endpoint values are common inputs. The coefficient bank for one feasible policy contains $\sum_{h=0}^H (h + 1)$ state vectors; across m policies this is $O(mSH^2)$, in addition to $O(mSH)$ policy indices. The implementation expands the chord one date at a time during certification, and charges that expansion to certificate time. These persistent-array counts exclude action-wide work buffers and the sparse kernel, and are not claims about process peak resident memory. If a row has at most z stored successors and A actions, an action-wide application costs $O(SAz)$. Naive repeated de Casteljau restriction of m_l local policy arrays to B cells costs $O(Bm_lSH^3)$; it can dominate when action optimization is cheap.

S.10.3. Proof of the Total Coefficient-Error Bound

Let V_n, L_n, Q_n be the primitive recursions in the main text. By backward induction, every fixed-policy value and every optimal value has absolute magnitude at most V_n . For any bounded continuation f with $\|f\|_\infty \leq V_{n+1}$,

$$\|T_{n,\lambda}^u f - T_{n,\lambda'}^u f\|_\infty \leq |\lambda - \lambda'|(\rho_n + \kappa_n V_{n+1}).$$

Monotonicity and the row-mass bound give a Lipschitz constant L_n for both optimal values and all fixed-policy values, whether or not the maximizing action changes.

A coefficient bound is needed in addition to this pointwise result. Let p^a be an endpoint-optimal Markov policy and let $F_{n,j}^{p^a,I}$ be its Bernstein coefficients on $I = [a, b]$. At date n , its conditional-count recursion is a convex combination of endpoint- a and endpoint- b backups of next-date coefficients. Relative to the pure- a policy value, induction and the preceding inequality give

$$\|F_{n,j}^{p^a,I} - J_n^{p^a,a}\|_\infty \leq \beta_n L_{n+1} w + (\rho_n + \kappa_n V_{n+1}) w = L_n w.$$

The same induction bounds the absolute coefficient magnitude by V_n . It is valid even though conditional counts change the regime probabilities: the current mixture weights are between zero and one, and the induction bound is uniform over the remaining count. Exact endpoint optimization gives $J_n^{p^a,a} = v_n^a$ at every state and date.

Next, $\|v_n^b - v_n^a\|_\infty \leq L_n w$, while the signed cross term obeys $\|D_n^u\|_\infty \leq \kappa_n L_{n+1} w^2$. The correction recursion therefore implies $\|M_n\|_\infty \leq Q_n w^2$. For $h \geq 2$, $0 \leq f_{h,j} \leq 1/2$. Consequently,

$$U_{n,j}^I - F_{n,j}^{p^a,I} \leq 2L_n w + \frac{1}{2} Q_n w^2.$$

For $h = 1$ the correction is zero, and at the terminal date the difference is zero. Selecting this one feasible policy already establishes the claimed bound; taking the better endpoint policy can only tighten it. Subdivision is positive and cannot invalidate the coefficient bound. Count dominance transfers the bound to the local count construction.

With $w \leq \epsilon/(4L_*)$ and $w \leq \sqrt{\epsilon/Q_*}$, the linear and quadratic terms are each at most $\epsilon/2$. A complete dyadic grid first reaching width w_* has fewer than $2/w_*$ intervals and at most one more anchor. This proves finite termination and the worst-case $O(\epsilon^{-1})$ statement for fixed primitive constants. The result does not require second-order smoothness of optimal values. For nonexact endpoints, their independently certified one-sided residual and policy-evaluation errors must be added before this termination rule is used.

S.10.4. *Scope of the Matched Frontier*

The full experiment reoptimizes every original R6 transition anchor on the unchanged target and independently reevaluates its policy coefficients. It uses the same endpoint-policy pair and eight closed cells on every compared interval. For $d = 0.5$, it also compares common coarser banks retaining every second or fourth original anchor, always including 1. These locations are inherited computational design choices, not newly timed autonomous anchor searches. Setup, anchor optimization, policy evaluation, local coefficient restriction, upper construction, and cell certification are charged to each method's reconstructed total. Shared work is measured once and attributed identically. Numerical ordering checks and replay diagnostics are validation overhead, not part of any arm's deployment workload. The results are one complete serial, single-thread pass per row on one machine; no cross-machine historical time is used as a comparator. The raw stage times and every interval bound are deposited.

S.11. PRIMITIVE ADJUSTMENT BOUNDS AND DECISION-CERTIFICATE REPLICATION

S.11.1. *The Primitive Option Bound*

For $z = u - 1 > 0$ and $a = \log c$,

$$U(c, u) = -\frac{e^{-za}}{z}, \quad U_u(c, u) = \frac{e^{-za}(1 + za)}{z^2}.$$

Differentiating again gives the strictly negative second derivative in the main text. For finite positive consumption support the expectation can be differentiated term by term. Taylor's integral formula and concavity imply, for $|\theta| \leq \bar{\theta}$,

$$g_\sigma \theta - \frac{1}{2} M_\sigma \theta^2 \leq F_\sigma(\theta) - F_\sigma(0) \leq g_\sigma \theta.$$

Subtracting $k\theta^2/2$, evaluating the lower quadratic at $\theta = g_\sigma/(k + M_\sigma)$ when this is feasible, and maximizing the upper quadratic on the whole line give the option bounds. The upper bound remains valid if the actual optimum is constrained. Subtracting the negative-class upper bound from the positive-class lower bound gives the relative-option inequality.

Since $D_A = A_+ - A_- > 0$ is a primitive constant in this specialization,

$$\Delta^0(d) = B_+ - B_- + dD_A, \quad \Delta^{\text{adj}}(d) = \Delta^0(d) + \Omega_+ - \Omega_-.$$

Both functions have the same strictly positive slope. Their unique indifference contracts therefore differ by $(\Omega_+ - \Omega_-)/D_A$. This proves the stated threshold shift and the intervening interval of opposite rankings. Uniqueness here follows from the explicitly constant duration difference in the two-stage model; it is not carried over to the full stopped portfolio model.

S.11.2. A Curvature Bound over an Entire Primitive Family

Curvature does not have to be estimated by sampling preference indices. In the notation above,

$$-U_{uu}(c, 1+z) = \frac{e^{-za}[(1+za)^2 + 1]}{z^3} = \int_a^\infty t^2 e^{-zt} dt.$$

Its second derivative with respect to z is $\int_a^\infty t^4 e^{-zt} dt \geq 0$. The integral is finite for every $z > 0$, also when $a < 0$. Hence the negative second derivative of expected utility is convex in z , and its maximum on the preference interval is attained at an endpoint. It is affine in the probability of a two-point consumption lottery. Its maximum over the stated probability/preference rectangle can therefore be bounded at its four vertices.

With $u_0 = 2$, $\bar{\theta} = 0.2$, $C_- = 0.5$, and $C_+ \in \{0.05, 0.8\}$, the positive-class shadow is affine and negative for every $p \in [0.06, 0.10]$. Its absolute value lies between 1.4820 and 3.1180; the absolute negative-class shadow is below 0.6138; and $M_+ < 17.342$. Feasibility of the lower-bound trial follows from $3.1180/20 < 0.2$. For every $k \in [20, 80]$,

$$\Omega_+ - \Omega_- \geq \frac{1.4820^2}{2(80 + 17.342)} - \frac{0.6138^2}{2 \cdot 20} > 0.00186.$$

The bound is conservative because it takes adverse cost values separately in its positive and negative terms. Division by $D_A = 0.5$ gives a reduction greater than 0.00372. This proves a statement on the entire probability/cost rectangle, not on the finite set of displayed optimizations.

TABLE S.7
PRIMITIVE PREDICTIONS AND OPTIMIZED ADJUSTMENT OPTIONS

p	k	Lower option	Actual option	Min. shift	Test d
0.06	20	0.0261621	0.0311074	0.0523241	0.723838
0.06	40	0.016882	0.0185198	0.0337641	0.733118
0.06	80	0.00973227	0.0102147	0.0194645	0.740268
0.08	20	0.0681238	0.0811337	0.136248	1.43188
0.08	40	0.0441692	0.0488284	0.0883384	1.45583
0.08	80	0.0257477	0.0271885	0.0514953	1.47425
0.10	20	0.12072	0.147117	0.24144	2.12928
0.10	40	0.0800377	0.0900327	0.160075	2.16996
0.10	80	0.0475675	0.0507768	0.095135	2.20243

Notes: Two-stage CRRA family: $u_0 = 2$, $|\theta| \leq 0.2$, $C_- = 0.5$, $C_+ \in \{0.05, 0.8\}$, and $A_+ - A_- = 0.5$. Option denotes $\Omega_+ - \Omega_-$. The primitive lower bound determines the test contract, at which every row has $\Delta^0 < 0 < \Delta^{\text{adj}}$. The continuous-family proof covers all $p \in [0.06, 0.10]$ and $k \in [20, 80]$, not just these rows. Values are in fixed cardinal utility and operating-benefit units.

The table compares the primitive lower bound with a one-dimensional strictly concave optimization in the same cardinal utility units. For each row the test contract is selected by $d = d^0 - \underline{\Omega}/(2D_A)$, before using optimized option values. It therefore tests a prediction from the bound rather than selecting a contract after locating a numerical reversal. The consumption values, probabilities, and duration difference specify an analytical family, not estimated parameters. In particular, the eight-date stopped economy retains its original state dynamics and is not replaced by this specialization.

S.11.3. *The Full-Model Joint-Region Calculation*

The risk certificate uses all 1,617 states, all eight decision dates, and the original 1,568-action target, comprising the common union mesh and the three frozen neural proposals. The declared initial state is $(u, X) = (2, 1.25)$, the adjustment cost is $k = 2$, the contract interval is $[0.4, 0.45]$, and the one-step shock-law mixture probability lies in $[0, 0.25]$. This region was selected during model exploration; it is not a preregistered or exhaustive characterization of all reversals.

For each adjustment regime and each of the four parameter corners, one backward optimization yields both first-risk classes: dates after the first use the regime's common feasible menu, and the two first-date maxima enforce positive and nonpositive risky positions separately. This is eight corner optimizations, yielding sixteen feasible class policies. The no-adjustment menu imposes zero deliberate adjustment at every date and state but retains preference shocks. Later portfolio signs remain unrestricted. Endpoint class values feed eight localized count constructions, one for each contract endpoint, class, and adjustment regime. Each corner policy is independently evaluated at reward contrasts zero and one, giving its Bernstein intercept and duration coefficients; this covers arbitrary intermediate d without a new policy solve.

Convexity of the optimal class value in d validates linear interpolation of its upper polynomials, not of its optimal policies. Feasible-policy values are affine in d exactly. The resulting degree-eight-by-degree-one tensor coefficients certify the entire rectangle. Subdivision is available if a sign interval contains zero; acceptance of the original closed rectangle in the deposited execution means no further subdivision is required. Pointwise replay at the corners checks implementations but is not the source of the continuum sign claim. All upper and lower tensor coefficients, policy indices, and policy-feature arrays are deposited, with a coefficient-only replay program that does not call an optimizer.

S.11.4. *Numerical Error Account*

The exact-arithmetic theorem is stated separately from its floating-point implementation. The application uses an a priori round-to-nearest allowance relative to the *stored finite kernel and reward arrays*. It is not a claim of directed-rounding interval arithmetic, and it does not include errors in choosing the state/action grid, constructing the quadrature model, or approximating a diffusion. Those omissions do not change which finite target the certificate concerns.

Every stored continuation row is nonnegative with at most sixteen pre-aggregation entries. Let $\hat{\beta}$ be the largest computed row sum plus 10^{-12} . The program verifies $\hat{\beta} \leq 1 + 2 \cdot 10^{-12}$. This margin dominates summation error for sixteen nonnegative entries of this size. Let R be twice the largest sum of the absolute stored base reward, duration reward at $|d| \leq 1$, and effort reward at $k = 2$, across both common kernels and every neural-proposal kernel. The use of $|d| \leq 1$ covers the intermediate feature calculation at $d = 1$, not merely the reported interval ending at 0.45. Let $G = 2\|g\|_\infty$ and

$$V = (G + HR) \max(1, \hat{\beta})^H, \quad q = 4096(H + 1), \quad \gamma_q = \frac{qu}{1 - qu}, \quad u = 2^{-53}.$$

The per-class allowance used in the calculation is

$$e = 2\gamma_q(1 + 2V + R) \max(1, \hat{\beta})^H. \quad (\text{S.3})$$

Here the operation budget is deliberately conservative. One continuation dot product requires at most sixteen products and fifteen additions; constructing its sparse row requires at most sixteen additions of original entries. Adding the three reward components, mixing endpoint laws, and mixing conditional-count branches takes fewer than forty additional scalar operations per path per date. An endpoint action maximum is nonexpansive in the sup norm: its error is bounded by the largest candidate error, not multiplied by the number of actions. Thus an endpoint solve, a selected-policy coefficient recursion, or a count recursion has fewer than $128H$ serial scalar operations along any contributing path. Computing intercept/duration features from values at zero and one adds one subtraction and one multiply-add per coefficient, with absolute magnitude bounded by $2V$. A degree- H restriction uses at most two triangular de Casteljau passes, with fewer than $6(H+1)^2$ scalar operations per retained coefficient path even if counted without sharing. For $H=8$, these counts, subsequent differences, and coefficient extrema are dominated by $4096(H+1)$. Positive mixing and row-mass stability propagate the absolute bound by at most the displayed mass factor. The factors two in R, G and outside γ_q cover input rounding margins and the addition of separately computed paths. The formula is an absolute-error allowance; signed reward cancellation is not analyzed with a relative-error claim. No overflow or underflow occurs in the recorded value range.

The same allowance is charged separately to the upper and feasible lower construction. A class-value gap therefore includes $2e$, and a class-difference sign interval also includes $2e$. The relative-option conclusion adds the already enlarged difference intervals. The information-relaxation and policy-bank errors are controlled by their computed coefficient gaps, not absorbed into an unexplained floating-point tolerance. Independent selected-kernel replay checks the actual arithmetic discrepancy, but small observed replay errors are not substituted for the a priori allowance. Supplementary JSON records the value and reward magnitudes, row-mass bound, operation budget, and final allowance.

S.11.5. *Reproduction and Preservation*

Run `bash replication/r7/run_all.sh` from the repository root, then compile `ECTA_R7.tex` and `SUPP_R7.tex`. The driver executes the dense independent checks, the primitive family, the joint-region certificate, the full matched upper comparison, and autonomous dyadic refinement. The full matched frontier reports one measured pass; the autonomous comparison starts each arm with a fresh policy cache and includes all rejected parent intervals. All retained R1–R6 numerical files are inputs or historical evidence, not silently regenerated by the R7 driver. A byte-level preservation check covers every inherited R6 file except the explicitly advanced revision index. A source inventory, environment record, complete stdout/stderr logs, generated tables, and compiled manuscript copies accompany the revision.

REFERENCES

- Alegre, Lucas Nunes, Ana L. C. Bazzan, and Bruno C. da Silva (2022): “Optimistic Linear Support and Successor Features as a Basis for Optimal Policy Transfer,” *Proceedings of Machine Learning Research*, 162, 394–413. []
- Amos, Brandon, Lei Xu, and J. Zico Kolter (2017): “Input Convex Neural Networks,” *Proceedings of Machine Learning Research*, 70, 146–155. [6]
- Barles, Guy, and Panagiotis E. Souganidis (1991): “Convergence of Approximation Schemes for Fully Nonlinear Second Order Equations,” *Asymptotic Analysis*, 4, 271–283. [5]

- Barreto, André, Will Dabney, Rémi Munos, Jonathan J. Hunt, Tom Schaul, Hado van Hasselt, and David Silver (2017): “Successor Features for Transfer in Reinforcement Learning,” *Advances in Neural Information Processing Systems*, 30. [17]
- Brown, David B., James E. Smith, and Peng Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58, 785–801. []
- Brumm, Johannes, and Simon Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85, 1575–1612. []
- Cai, Wei, Shuixin Fang, and Tao Zhou (2025): “SOC-MartNet: A Martingale Neural Network for the Hamilton–Jacobi–Bellman Equation without Explicit $\inf H$ in Stochastic Optimal Controls,” arXiv:2405.03169v4, March 15. []
- Chang, Hyeon Soo (2021): “On-line Policy Iteration with Policy Switching for Markov Decision Processes,” arXiv:2112.02177. [17]
- Csáji, Balázs Csanád, and László Monostori (2008): “Value Function Based Reinforcement Learning in Changing Markovian Environments,” *Journal of Machine Learning Research*, 9, 1679–1709. []
- Duffie, Darrell, and Larry G. Epstein (1992): “Stochastic Differential Utility,” *Econometrica*, 60, 353–394. [10]
- Epstein, Larry G., and Stanley E. Zin (1989): “Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework,” *Econometrica*, 57, 937–969. []
- Han, Jiequn, and Weinan E (2016): “Deep Learning Approximation for Stochastic Control Problems,” arXiv:1611.07422. []
- Han, Jiequn, Arnulf Jentzen, and Weinan E (2018): “Solving High-Dimensional Partial Differential Equations Using Deep Learning,” *Proceedings of the National Academy of Sciences*, 115, 8505–8510. []
- Hornik, Kurt, Maxwell Stinchcombe, and Halbert White (1990): “Universal Approximation of an Unknown Mapping and Its Derivatives Using Multilayer Feedforward Networks,” *Neural Networks*, 3, 551–560. [5]
- Jacka, Saul D., and Aleksandar Mijatović (2015): “On the Policy Improvement Algorithm in Continuous Time,” arXiv:1509.09041. [2]
- Judd, Kenneth L. (1998): *Numerical Methods in Economics*. Cambridge, MA: MIT Press. []
- Kim, Yeongjong, Yeoneung Kim, Minseok Kim, and Namkyeong Cho (2025): “Neural Policy Iteration for Stochastic Optimal Control: A Physics-Informed Approach,” arXiv:2508.01718v1, August 3. []
- Kim, Yeongjong, Minseok Kim, Yeoneung Kim, and Namkyeong Cho (2026): “Physics-Informed Policy Iteration for High-Dimensional Hamilton–Jacobi–Bellman Equations: Interior Error Bounds without Boundary Data,” arXiv:2508.01718v2, August 9. []
- Lee, Jae Yong, and Yeoneung Kim (2024): “Hamilton–Jacobi Based Policy-Iteration via Deep Operator Learning,” arXiv:2406.10920. []
- Merton, Robert C. (1971): “Optimum Consumption and Portfolio Rules in a Continuous-Time Model,” *Journal of Economic Theory*, 3, 373–413. []
- Milgrom, Paul, and Ilya Segal (2002): “Envelope Theorems for Arbitrary Choice Sets,” *Econometrica*, 70, 583–601.
- Nemecsek, Mark, and Ronald Parr (2021): “Policy Caches with Successor Features,” *Proceedings of Machine Learning Research*, 139, 8025–8033. []
- Pearlmutter, Barak A. (1994): “Fast Exact Multiplication by the Hessian,” *Neural Computation*, 6, 147–160. [17]
- Qi, Qian (2024): “Neural Foundations,” working paper, SSRN 5072524. [6]
- Quatmann, Tim, Christian Dehnert, Nils Jansen, Sebastian Junges, and Joost-Pieter Katoen (2016): “Parameter Synthesis for Markov Models: Faster Than Ever,” in *Automated Technology for Verification and Analysis*, Lecture Notes in Computer Science, 9938, 50–67. []
- Raissi, Maziar, Paris Perdikaris, and George Em Karniadakis (2019): “Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations,” *Journal of Computational Physics*, 378, 686–707. []
- Stokey, Nancy L., Robert E. Lucas, Jr., and Edward C. Prescott (1989): *Recursive Methods in Economic Dynamics*. Cambridge, MA: Harvard University Press. []